

Test Coverage Report

Smart Contract: MultiSigWallet

Course: CS 218 Programmable and Interoperable Blockchains

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Team: 412

1. Execution Summary

The Foundry test suite was executed successfully to validate the security and functionality of the MultiSigWallet smart contract. The execution completed rapidly, demonstrating optimal local testing performance.

- **Total Tests Run:** 30
- **Passed:** 30
- **Failed / Skipped:** 0
- **Execution Time:** 54.54ms (8.73ms CPU time)

2. Coverage Metrics

The table below details the code coverage achieved across the repository. The core logic contract achieves perfect coverage, securing all conditional pathways.

3. Analysis

- **Core Logic:** The primary MultiSigWallet.sol contract demonstrates a robust security posture with **100% coverage** across all tracking metrics.
- **Branch Integrity:** Achieving 100% branch coverage (11/11 branches) ensures that all **require** statements, access controls, threshold evaluations, and edge cases have been rigorously tested and validated.

File	% Lines	% State- ments	% Branches	% Funcs
script/deploy.s.sol	0.00% (0/16)	0.00% (0/21)	100.00% (0/0)	0.00% (0/1)
src/MultiSigWallet.sol	100.00% (53/53)	100.00% (49/49)	100.00% (11/11)	100.00% (13/13)
test/MultiSigWallet.t.sol	100.00% (5/5)	100.00% (3/3)	100.00% (0/0)	100.00% (2/2)
Total	78.38% (58/74)	71.23% (52/73)	100.00% (11/11)	93.75% (15/16)

- **Deployment Script:** The 0% coverage for `deploy.s.sol` is standard industry practice. Foundry scripts dictate off-chain deployment steps and state setup rather than on-chain logic, meaning they do not require unit testing in this context.